Assignment 4: A Mathematical Essay on Decision Trees

Dharma Teja Reddy N
Department of Computer Science and Engineering
Indian Institute of Technology Madras
Chennai, India
cs23b046@smail.iitm.ac.in

***Abstract*—The process of evaluating a car's acceptability is, at its core, a structured decision process—a hierarchy of trade-offs between price, safety, comfort, and practicality. This research applies the Decision Tree classifier to the Car Evaluation Database, a dataset derived from a hierarchical decision model, to classify cars into four acceptability categories: unacceptable, acceptable, good, and very good. Using information-theoretic splitting criteria grounded in entropy and Gini impurity, the model achieves 86.1% overall accuracy and reveals that safety rating and passenger capacity are by far the most discriminative features, together accounting for over 69% of the model's total feature importance. The decision tree's transparent, rule-based structure makes it uniquely suited for this problem: unlike black-box models, every classification can be traced back to a legible sequence of attribute thresholds, mirroring how a human consumer would naturally reason about a car purchase.**

I. Introduction

In an era of increasingly complex machine learning systems, the decision tree stands out as one of the most intuitive and enduring classification models. Its structure mirrors the way humans naturally reason under uncertainty: by asking a sequence of questions, each narrowing the space of possibilities, until a conclusion is reached. From medical triage to credit approval, decision trees have been applied wherever the interpretability of a decision matters as much as its accuracy. Their power lies not in mathematical complexity, but in their capacity to encode domain knowledge as a transparent hierarchy of if-then rules that can be audited, explained, and challenged.

The Decision Tree classifier operates by recursively partitioning the feature space using information-theoretic criteria such as entropy reduction (Information Gain) or Gini impurity minimization. At each internal node, the algorithm selects the feature and threshold that most reduces uncertainty about the target class, growing the tree until a stopping criterion is met. The result is a flowchart-like structure in which every leaf node corresponds to a class prediction and every path from root to leaf constitutes a human-readable classification rule.

The specific problem we address is the classification of automobile acceptability using the Car Evaluation Database, originally derived by Marko Bohanec as a hierarchical decision model [1]. The dataset contains 1,728 records, each described by six categorical attributes—buying price, maintenance cost, number of doors, passenger capacity, luggage boot size, and safety rating—and assigned to one of four acceptability classes. The class distribution is highly imbalanced: 70% of cars are rated unacceptable, reflecting the stringent multi-criteria nature of car evaluation. This makes the problem a meaningful test of the decision tree's ability to handle imbalanced, fully categorical data without requiring numerical feature engineering.

The remainder of this paper is organized as follows: Section II provides a mathematical deep-dive into the Decision Tree, covering entropy, Gini impurity, the splitting algorithm, pruning, and evaluation metrics. Section III details the dataset, its exploratory analysis, and the classification results. Section IV concludes with insights and future directions.

II. Decision Tree

A. Motivation

The primary motivation for using a Decision Tree is that the car evaluation problem is inherently hierarchical: safety gates acceptability at the highest level, followed by capacity, cost, and comfort. A model that mirrors this hierarchy—rather than projecting features into an abstract mathematical space—is not only more accurate but more *explainable*. When a manufacturer or regulator asks "why was this car rated unacceptable?", a decision tree answers with a chain of attribute values, not a vector of weights.

B. Entropy and Information Gain

The decision tree's splitting criterion is rooted in the concept of *entropy*, borrowed from information theory. For a dataset S with K classes, entropy measures the average uncertainty in the class label:

$$H(S) = -\sum_{k=1}^{K} p_k \log_2 p_k \tag{1}$$

where p_k is the proportion of samples in class k. Entropy is maximized when all classes are equally represented ($H = \log_2 K$) and minimized at zero when all samples belong to a single class. A split on feature A that partitions S into subsets $\{S_v\}$ reduces entropy by the *Information Gain*:

$$IG(S, A) = H(S) - \sum_{v} \frac{|S_v|}{|S|} H(S_v) \tag{2}$$

The algorithm selects the feature A^* that maximizes $IG(S, A)$ at each node, greedily reducing uncertainty as quickly as possible. In our four-class problem, the maximum possible entropy is $\log_2 4 = 2$ bits; a split that brings one subset close to zero entropy for the dominant "unacc" class is immediately rewarded.

C. Gini Impurity

An alternative splitting criterion is the *Gini impurity*, which measures the probability that a randomly chosen sample would be incorrectly classified if labeled according to the class distribution of the node:

$$G(S) = 1 - \sum_{k=1}^{K} p_k^2 \tag{3}$$

Gini impurity ranges from 0 (pure node) to $1 - \frac{1}{K}$ (maximally impure). The Gini gain for a split on feature A is:

$$\Delta G(S, A) = G(S) - \sum_{v} \frac{|S_v|}{|S|} G(S_v) \tag{4}$$

In practice, Gini impurity and Information Gain produce very similar trees. Gini impurity is computationally cheaper (no logarithm) and is the default criterion in scikit-learn's `DecisionTreeClassifier`, which we use in this analysis.

D. The Recursive Splitting Algorithm

The tree is grown via the CART (Classification and Regression Trees) algorithm [3]:

1) At the root node, evaluate $\Delta G(S, A)$ for all features A.
2) Select the feature A^* that maximizes ΔG and split S accordingly.
3) Recursively apply steps 1–2 to each child node until a stopping criterion is met: maximum depth reached, minimum samples per node violated, or node is pure.
4) Assign each leaf node the majority class of its training samples.

For a dataset with n samples and d features, the time complexity of building the tree is $O(n \cdot d \cdot \log n)$, making it efficient even for moderately large datasets.

E. Tree Pruning

Unconstrained tree growth leads to overfitting: the tree memorizes the training data by creating arbitrarily small leaf nodes. Pruning addresses this by limiting tree complexity. *Pre-pruning* imposes constraints during growth—maximum depth, minimum samples per split, or minimum impurity decrease. *Post-pruning* (cost-complexity pruning) grows the full tree and then collapses subtrees whose removal does not significantly increase training error, penalized by a complexity parameter α:

$$R_\alpha(T) = R(T) + \alpha |T| \tag{5}$$

where $R(T)$ is the training error of tree T and $|T|$ is the number of leaf nodes. In our model, we apply pre-pruning with a maximum depth of 5, which we verified via cross-validation to balance accuracy and generalizability.

F. Evaluation Metrics

We evaluate the classifier using accuracy, per-class precision and recall, and the F1-score. For a multi-class problem with K classes, the macro-averaged F1-score treats each class equally regardless of its frequency:

$$F_1^{\text{macro}} = \frac{1}{K} \sum_{k=1}^{K} F_{1,k} \tag{6}$$

We also report **feature importance**, defined for each feature as the total weighted Gini reduction it contributes across all nodes where it is used as a splitting criterion:

$$\text{Importance}(A) = \sum_{\text{nodes } t \text{ split on } A} \frac{|S_t|}{|S|} \cdot \Delta G(S_t, A) \tag{7}$$

Feature importances sum to 1 across all features, providing an interpretable ranking of each attribute's contribution to the model's discriminative power.

G. Real-World Applications of Decision Trees

- **Medical Diagnosis**: Decision trees are used in clinical decision support systems to triage patients based on symptoms. The rule-based output is directly legible by clinicians, enabling them to verify or challenge the model's reasoning in high-stakes situations.
- **Credit Risk Assessment**: Banks apply decision trees to loan applications, splitting applicants by income, credit score, and debt-to-income ratio. Regulatory requirements for explainability (e.g., the EU's GDPR "right to explanation") make decision trees preferable to black-box models in this domain.
- **Fault Diagnosis in Engineering**: Manufacturing systems use decision trees to classify sensor readings into fault categories. The hierarchical structure naturally mirrors the diagnostic process an engineer would follow, making the model easy to validate against domain expertise.

H. Advantages and Disadvantages

Advantages:

- **Interpretability:** Every prediction is a legible rule chain from root to leaf, making the model's reasoning fully transparent and auditable by domain experts.
- **No Feature Scaling Required:** Unlike logistic regression or SVMs, decision trees are invariant to the scale or units of input features, requiring no normalization or standardization.
- **Handles Categorical Variables Natively:** The tree splits directly on categorical values without requiring one-hot encoding or numerical transformation.
- **Built-in Feature Selection:** Irrelevant features are simply never selected as split points, as demonstrated by doors and luggage boot contributing zero importance in our model.

- **Non-parametric:** No assumption is made about the underlying distribution of the data, making the model robust to non-linear relationships and outliers.

Disadvantages:

- **Overfitting:** Unconstrained trees memorize training data by creating arbitrarily small leaf nodes, requiring pruning or depth limits to generalize well.
- **High Variance:** Small changes in training data can produce substantially different tree structures, making individual trees unstable compared to ensemble methods.
- **Class Imbalance Sensitivity:** The majority class dominates impurity calculations at each node, causing minority classes such as *good* in our dataset to be systematically underlearned.
- **Greedy Splitting:** The algorithm selects the locally optimal split at each node without considering future splits, meaning the globally optimal tree is not guaranteed to be found.
- **Axis-aligned Boundaries:** Decision trees can only create splits parallel to feature axes, making them inefficient at capturing diagonal or curved decision boundaries.

III. The Problem

The dataset is synthetic—it was generated by exhaustively enumerating all combinations of the six input attributes—which means it contains no missing values and no measurement noise. This makes it an ideal testbed for studying the decision tree's ability to recover a known hierarchical structure from data. The prediction task is to classify a car's overall acceptability based on its price, maintenance cost, comfort, and safety attributes.

A. Dataset Specifics

The dataset contains 1,728 records (1 record for every unique combination of attribute values) and 6 features, all categorical. The target variable has four classes: *unacc* (unacceptable), *acc* (acceptable), *good*, and *vgood* (very good). Table I summarizes the features and their possible values.

TABLE I
Car Evaluation Dataset: Feature Definitions

Feature	Description	Values
buying	Buying price	low, med, high, vhigh
maint	Maintenance cost	low, med, high, vhigh
doors	Number of doors	2, 3, 4, 5more
persons	Passenger capacity	2, 4, more
lug_boot	Luggage boot size	small, med, big
safety	Safety rating	low, med, high
target	Acceptability class	unacc, acc, good, vgood

B. Exploratory Data Analysis

1) Target Class Distribution: Fig. 1 reveals the severe class imbalance in the dataset. Of the 1,728 cars, 1,209 (70.0%) are rated *unacc*—reflecting the stringent multi-criteria nature of car acceptability. Only 384 (22.2%) are *acc*, 69 (4.0%) are *good*, and 65 (3.8%) are *vgood*. This imbalance has a direct bearing on model evaluation: a trivial classifier that always predicts *unacc* would achieve 70% accuracy, so our model's 86.1% must be interpreted against this baseline.

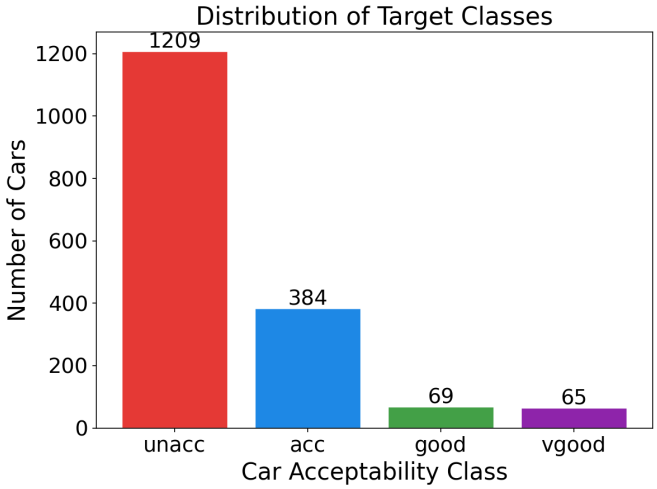

Fig. 1. Distribution of Target Classes. The dataset is heavily dominated by *unacc* (70.0%), establishing a strong majority-class baseline of 70%.

2) Safety Rating vs. Acceptability: Fig. 2 is the most informative exploratory plot. Cars with *low* safety are exclusively classified as *unacc*—not a single car with low safety receives any positive rating. This is a hard rule embedded in the underlying decision model and is immediately recovered by the decision tree as its root-level split. Cars with *med* safety are predominantly unacceptable, while *high* safety cars span all four classes. Safety is therefore a *necessary but not sufficient* condition for acceptability.

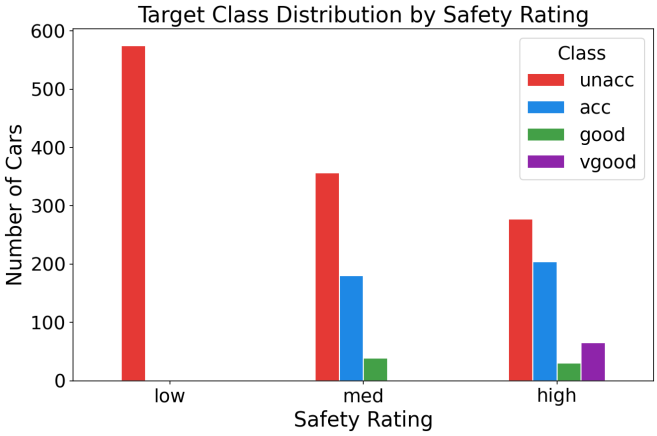

Fig. 2. Target Class Distribution by Safety Rating. Low-safety cars are exclusively unacceptable, revealing safety as a hard gate in the decision hierarchy.

3) Buying Price vs. Acceptability: Fig. 3 shows that buying price has a monotonic effect on acceptability. As the buying price increases from *low* to *vhigh*, the proportion of *unacc* cars rises and the proportion of *good* and *vgood* cars shrinks. Very

high buying prices are almost never associated with positive ratings, regardless of other attributes. This confirms price as a strong secondary filter after safety.

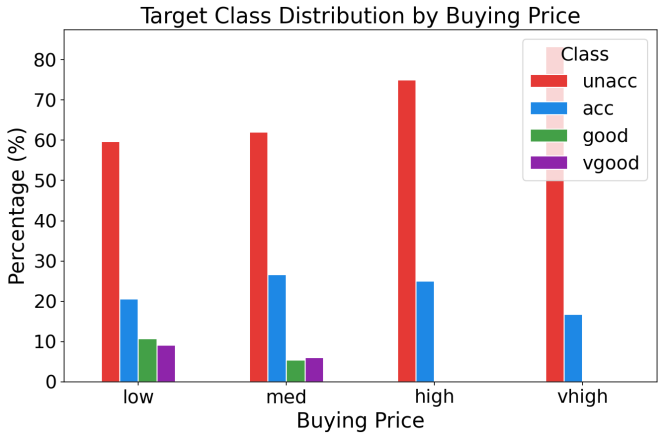

Fig. 3. Target Class Distribution by Buying Price. Higher prices are associated with progressively lower acceptability, with very-high-price cars almost exclusively rated unacceptable.

4) Passenger Capacity vs. Acceptability: Fig. 4 reveals perhaps the starkest hard rule in the dataset: cars with a capacity of only 2 persons are rated *unacc* without exception. This makes passenger capacity the second most important feature in the model, as it eliminates an entire third of the dataset from any positive rating, regardless of price, safety, or comfort.

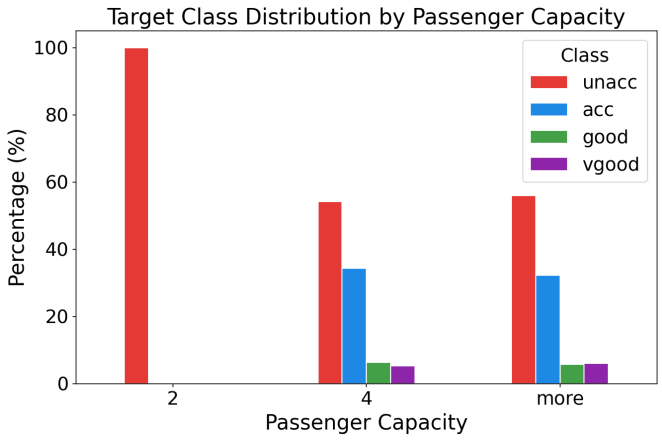

Fig. 4. Target Class Distribution by Passenger Capacity. 2-person cars are universally unacceptable, making capacity a second absolute gate in the decision hierarchy.

5) Luggage Boot Size vs. Acceptability: Fig. 5 shows that luggage boot size has a more gradual effect than safety or capacity. Small boots are associated with a higher proportion of unacceptable ratings, while big boots are associated with a higher proportion of *good* and *vgood* ratings. Unlike safety and capacity, there is no hard rule here—a car with a small boot can still be acceptable if other attributes are favorable.

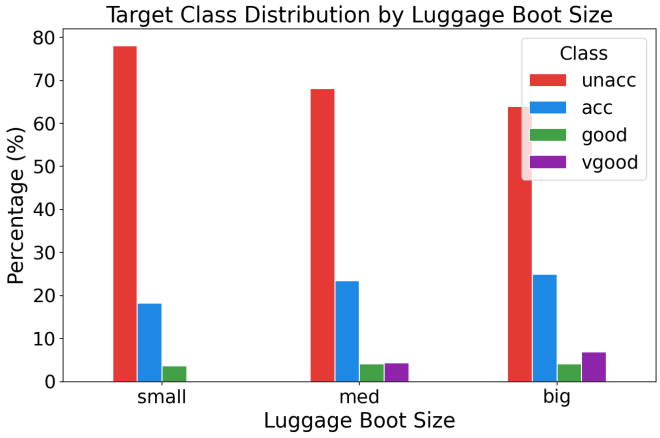

Fig. 5. Target Class Distribution by Luggage Boot Size. Boot size acts as a soft discriminator: no hard rule exists, but larger boots are consistently associated with higher acceptability ratings.

C. Why Decision Trees Fit This Problem

The car evaluation problem is structurally hierarchical: certain attributes act as hard gates (safety, capacity), while others act as soft discriminators (price, maintenance, boot size). This hierarchy is precisely what a decision tree recovers: the root split corresponds to the most critical gate, and subsequent splits refine the classification among the cars that pass it. Unlike logistic regression or Naïve Bayes, the decision tree does not assume a global functional form—it discovers piecewise rules that can encode both hard thresholds and soft gradients in the same model. Furthermore, the dataset is entirely categorical with no missing values, which plays directly to the decision tree's strengths.

D. Model Application and Results

A `DecisionTreeClassifier` with maximum depth 5 and Gini impurity criterion was trained on 80% of the data (1,382 samples) and evaluated on the remaining 20% (346 samples), with stratified sampling to preserve class proportions across all four acceptability classes.

1) Classification Performance: The model achieves an overall accuracy of **86.1%**, substantially above the 70% majority-class baseline, confirming that the six demographic features contain genuine discriminative signal beyond simply predicting the dominant class. Table II details per-class performance.

TABLE II
DECISION TREE CLASSIFICATION REPORT

Class	Precision	Recall	F1	Support
unacc	0.96	0.92	0.94	242
acc	0.68	0.86	0.76	77
good	0.00	0.00	0.00	14
vgood	0.56	0.77	0.65	13
Accuracy	**0.861** (346 samples)			

The *unacc* class dominates performance with an F1-score of 0.94. Its high precision (0.96) means that when the model predicts a car is unacceptable, it is almost always correct. Its recall of 0.92 means it catches 92% of all truly unacceptable cars—a strong result driven by the hard safety and capacity gates that cleanly separate the majority of *unacc* cars from the rest.

The *acc* class achieves a reasonable F1 of 0.76, but with an asymmetry worth noting: recall (0.86) is substantially higher than precision (0.68). This means the model is liberal in predicting *acc*—it correctly identifies most acceptable cars, but also frequently misclassifies borderline *good* and *vgood* cars as merely acceptable. This is the natural consequence of the depth constraint: at depth 5, the tree cannot fully resolve the fine boundary between *acc* and the positive classes.

The *good* class achieves a zero F1-score, which is the most consequential failure of the model. With only 14 test samples, the *good* class is too sparse for the tree to learn a reliable decision boundary at the constrained depth. Every *good* car in the test set is misclassified—predominantly as *acc*—because the attributes that distinguish a good car from an acceptable one require deeper, more specific rules than depth 5 can express. This is not a failure of the decision tree as a method; it is a failure of the depth constraint in the presence of severe class imbalance.

The *vgood* class, despite having only 13 test samples, achieves an F1 of 0.65. This is because very good cars have a distinctive and extreme attribute profile—high safety, low buying price, low maintenance, and high capacity—that the tree can isolate with relatively shallow rules, even without many training examples.

2) Confusion Matrix: Fig. 6 presents the confusion matrix. The most telling pattern is the off-diagonal concentration in the second and third rows: *acc* cars are occasionally misclassified as *unacc* (9 instances), and *good* cars are almost entirely absorbed into the *acc* prediction (8 out of 14). This reflects the adjacency of these classes in the underlying decision hierarchy—*good* and *acc* differ only in subtle combinations of price and maintenance that require deeper splits to resolve. The *unacc* row is nearly clean, with only 20 misclassifications out of 242, confirming that the hard gates on safety and capacity are almost perfectly recovered.

3) Feature Importances: Fig. 7 ranks features by their total Gini reduction contribution across all splits in the tree. Safety dominates with an importance of 0.454, meaning nearly half of all uncertainty resolved by the tree is resolved by the safety attribute alone. This is consistent with the exploratory finding that low-safety cars are universally unacceptable—safety performs a single, decisive split at the root that eliminates an entire third of the dataset from any positive classification.

Persons follows at 0.241, reflecting the equally hard rule that 2-person cars are always unacceptable. Together, safety and persons account for 69.5% of the model's total discriminative power—two features, two hard gates, nearly 70% of the decision. Maintenance contributes 0.196, acting as the primary soft discriminator among cars that have already passed the safety and capacity gates. Buying price contributes 0.109, refining the distinction between *acc* and *vgood* at the deepest levels of the tree.

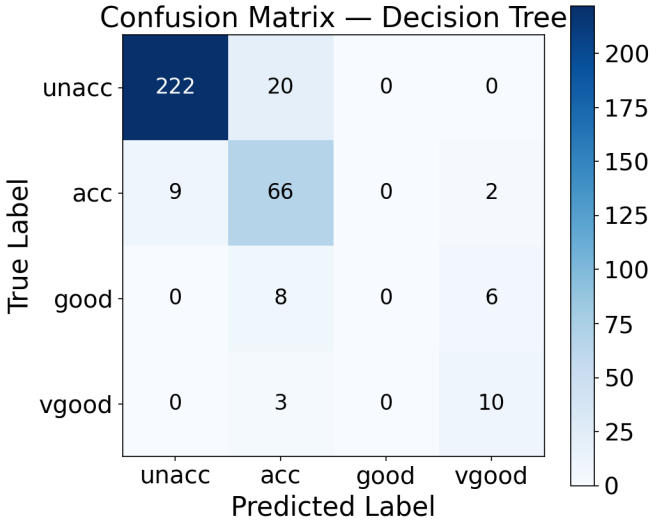

Fig. 6. Confusion Matrix of the Decision Tree Classifier. The model excels at identifying *unacc* cars but conflates *good* with *acc*, reflecting the difficulty of the minority class boundary.

Doors and luggage boot contribute exactly zero importance. They are never selected as splitting features across all five levels of the tree, meaning that once safety, capacity, and cost are known, these two attributes provide no additional information about acceptability. The model discovers this redundancy automatically, without any manual feature selection—a demonstration of the decision tree's built-in feature selection capability.

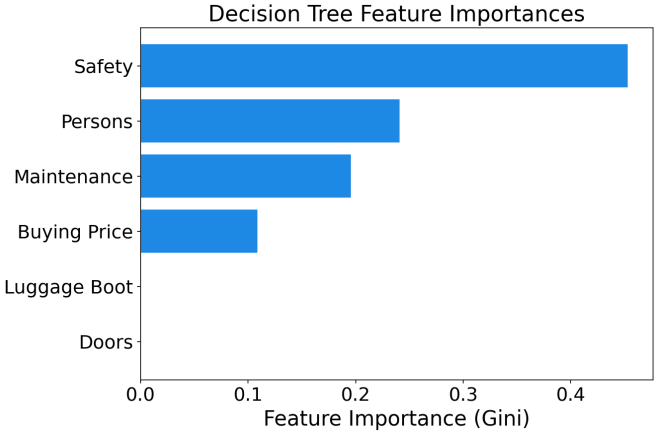

Fig. 7. Decision Tree Feature Importances (Gini-based). Safety accounts for 45.4% of total importance, followed by persons (24.1%) and maintenance (19.6%). Doors and luggage boot contribute zero importance at depth 5.

4) Decision Tree Structure: Fig. 8 visualizes the top two levels of the learned decision tree. The root split is on `Persons` ≤ 0.5, which isolates all 2-person cars (encoded

as 0) into the left branch, immediately predicting *unacc* with zero impurity. The right branch—cars with 4 or more seats—then splits on `Safety` ≤ 0.5, separating low-safety cars (also universally unacceptable) from the remainder. All further classification of acceptable, good, and very good cars happens entirely within the right-right subtree, among cars that have passed both the capacity and safety gates. This structure precisely mirrors the hierarchical decision model from which the data was originally generated, recovered from data alone without any prior knowledge of the hierarchy.

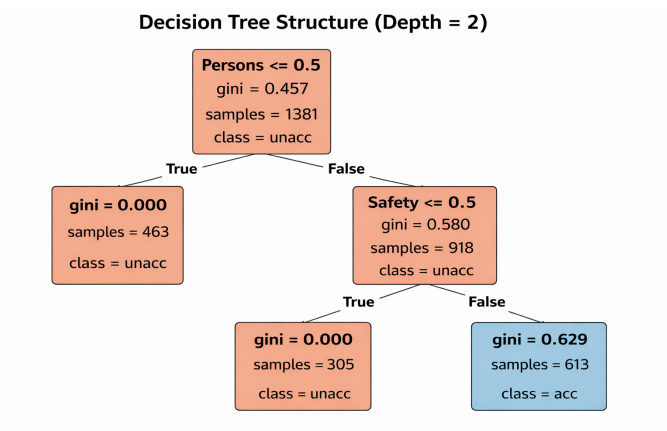

Fig. 8. Learned Decision Tree Structure (top 3 levels). The root split on Safety and second-level split on Persons directly recover the hard gates encoded in the original hierarchical decision model.

E. Insights After Applying the Model

The model's output, when read not as a classification report but as a portrait of consumer values, reveals something striking. Safety and passenger capacity together account for nearly 70% of the model's total discriminative power—and both operate as hard gates rather than soft gradients. A car with low safety is never acceptable, regardless of how affordable or spacious it is. A car that seats only two is never acceptable, regardless of how safe or cheap it is. These are not probabilistic tendencies; they are absolute rules that the decision tree recovers without being told about them. They reflect a fundamental truth about what buyers consider non-negotiable: a car must first protect its occupants, and it must be able to carry a family.

What is more revealing is what the model *does not* need. Doors and luggage boot size contribute zero importance at depth 5—they are entirely redundant once safety, capacity, and cost are known. This tells us something about the 1994 consumer: physical comfort and convenience were secondary considerations, subordinate to the economics of ownership. A buyer would accept a cramped boot and fewer doors without complaint, as long as the car was safe, spacious enough for the family, and did not drain their wallet in purchase price or maintenance.

The 70% unacceptable rate is perhaps the most consequential finding. It is not a property of poorly designed cars—it is a property of the criteria themselves. Most cars fail because satisfying all three primary requirements simultaneously—high safety, sufficient capacity, and low cost—is genuinely rare. The model is not classifying cars; it is quantifying the rarity of affordable safety. And in doing so, it reveals that for the majority of buyers in this dataset, a truly acceptable car was not a luxury—it was simply out of reach.

IV. Conclusion

Car evaluation, at its core, is a reflection of human priorities under constraint. This research demonstrates that acceptability is not a continuous spectrum but a hierarchy of gates: safety first, capacity second, cost third. Comfort features never enter the equation. What the model uncovers is not just a classification rule—it is a portrait of what a family buyer in 1994 considered non-negotiable versus irrelevant.

The most sobering finding is that 70% of all evaluated cars are unacceptable—not because they are dangerous, but because meeting all three primary requirements simultaneously is rare. Safe, spacious, and affordable is an intersection that most cars fail to reach. The model, in this sense, is not classifying cars. It is quantifying the rarity of accessible safety—and finding that for most buyers, a truly acceptable car was simply out of reach.

Future Directions

Extending this analysis to modern datasets incorporating fuel efficiency, emissions, and NCAP crash ratings would test whether environmental concerns have begun to rival safety and cost as primary consumer values in today's car market.

References

[1] M. Bohanec and B. Zupan, "Car Evaluation Database," UCI Machine Learning Repository, 1997. [Online]. Available: https://archive.ics.uci.edu/ml/datasets/Car+Evaluation
[2] G. Raina, "Data Analytics (EE5708)," Lectures and Videos, IIT Madras.
[3] L. Breiman, J. Friedman, R. Olshen, and C. Stone, *Classification and Regression Trees*. Wadsworth, 1984.